AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Gene expression and genetics

Every cell in your body carries the same DNA, yet a nerve cell and a liver cell look and behave very differently. This page explains how cells switch genes on and off, how one gene can produce more than one protein, how chemical changes to DNA and its packaging alter gene activity without changing the base sequence, and why the characteristics we observe depend on both genes and the environment.

1. Controlling gene expression

Gene expression means turning the information in a gene into a working product. For a protein-coding gene this happens in two stages. In transcription, a section of DNA is copied into a molecule of messenger RNA (mRNA) inside the nucleus. In translation, the mRNA travels to a ribosome in the cytoplasm and is read in codons to build a polypeptide chain. Both stages can be regulated, but the commonest point of control is transcription, because switching a gene off at this stage stops the whole chain of events before any protein is made.

Different cells express different combinations of genes. A cell that makes large amounts of digestive enzymes has many copies of the enzyme mRNA, while a muscle cell is rich in the proteins that contract. Because the proteins present in a cell, and how much of each, largely set its structure and function, the cell's type is effectively decided by which genes are active. Every cell has the full set of genes, so specialisation is about selection, not loss.

- **Transcription:** copying one strand of a gene's DNA into pre-mRNA, catalysed by RNA polymerase.
- **Translation:** reading mRNA codons at a ribosome to join amino acids in the order specified.
- **Transcription factor:** a protein that binds to DNA near a gene and either helps or blocks RNA polymerase from starting transcription.
- **Cell specialisation:** the permanent change in cell structure and function that follows from switching on a particular set of genes.

Transcription factors

The most common way a cell controls a gene is by switching its transcription on or off. Transcription factors do this. Some bind to the promoter region, the stretch of DNA just before a gene where RNA polymerase normally attaches, and either recruit the polymerase or prevent it from binding. Others bind to enhancer sequences, which may lie some distance from the gene. These enhancer-binding factors change the local structure of chromatin, the DNA-protein complex in the nucleus, so that the gene is either more or less accessible to the polymerase.

Open chromatin, where the DNA is loosely packed, is linked to active genes. Closed or condensed chromatin is linked to inactive genes. Controlling transcription is an efficient strategy: one mRNA molecule can be translated many times by ribosomes, so a single switch at the mRNA stage produces a large number of protein molecules.

KEY IDEA

Transcription is the most efficient control point because one mRNA molecule can be read by many ribosomes at once. Switching transcription off therefore cuts protein production far more than blocking a single translation event would.

Cell specialisation in practice

Chemical signals such as hormones trigger specialisation. A hormone binds to a receptor, the hormone-receptor complex moves into the nucleus, and it activates particular genes. Those genes are transcribed into mRNA, the mRNA is translated, and the new protein changes what the cell does. Because the change is carried by proteins that persist, and because the cell passes its gene-expression pattern to daughter cells at division, the specialised state is maintained over time.

EXAM TIP

In exam answers, describe specialisation as a chain: signal, receptor, activated gene, transcription, translation, protein, changed cell. Examiners look for the order of steps, not just the words "gene switched on".

2. One gene, many proteins

If each gene made exactly one protein, the number of proteins in a cell would equal the number of genes. In reality a single gene can give rise to several different proteins. The explanation lies in what happens to the RNA transcript after transcription, not in the DNA itself.

How one gene gives different mRNAs

1. Transcription of the gene produces a primary RNA transcript (pre-mRNA) that contains both exons and introns. Exons are the sequences that will end up in the final message; introns are the sequences that will be removed.

2. Splicing removes the introns from the transcript. Spliceosomes, which are complexes of proteins and small RNAs, carry out this cutting and rejoining.

   Only the exons carry codons for amino acids, so the introns must be removed before translation can produce a correct polypeptide.

3. The exons are not always joined in the same combination. In some cells certain exons are left out, and in others different exons are kept. This is alternative splicing.

4. Each combination gives a different mRNA, with a different sequence and length. A longer mRNA contains more codons, so the polypeptide it encodes has more amino acids.

5. Different amino acid sequences form different bonds during folding, so the same gene can produce proteins that fold into different three-dimensional shapes and perform different functions.

Shape, not sequence alone, determines what a protein does, so changing which exons are joined changes function.

COMMON MISTAKE

Alternative splicing does not change the DNA. Students often write that the gene "mutates" or "changes" to make new proteins. The gene sequence stays the same; only the processing of its transcript differs.

3. Epigenetics: control beyond the base sequence

Epigenetics is the study of control over gene activity that does not come from the order of bases in the DNA. The same base sequence can be active in one cell and silent in another. Two main systems do this: DNA methylation and histone modification. Both respond to the environment, and both can be inherited when cells divide, which is why they matter for development and for the phenotype of an organism.

RNA splicing also counts as an epigenetic process in this sense, because it changes which mRNAs and proteins are produced from the same genetic code.

DNA methylation

A methyl group (-CH3) can be added to a cytosine base when that cytosine sits next to a guanine in the DNA chain, with a phosphate linking them. The enzyme that adds it is a DNA methyltransferase, which uses a methyl donor molecule. Once a methyl group is attached, the local DNA is less available to the transcription machinery, so the gene tends to be silenced. Removing the methyl group, which is demethylation, allows the gene to be transcribed again.

Methylation also changes the histones that the DNA wraps around, so one chemical change can affect gene activity in more than one way.

Histone modification

DNA is wound around histone proteins, which are rich in positively charged amino acids such as lysine. The DNA is negatively charged because of its phosphate groups, so the two are attracted to each other. The DNA-histone complex is chromatin, and chromatin packs into chromosomes. How tightly the DNA is wound determines whether genes can be read. Densely packed heterochromatin hides genes from the transcription machinery. Loosely packed euchromatin exposes them.

Chemical groups added to histones change how tightly they hold the DNA:

- **Acetylation:** an acetyl group is added to a lysine on the histone. The acetyl group neutralises the positive charge, so the histone holds DNA less tightly. The chromatin opens and genes in that region become available for transcription.
- **Methylation (of histones):** a methyl group is added to a lysine. This is usually linked to gene silencing, and sometimes to whole-chromosome silencing, because the nucleosomes pack more tightly and transcription factors cannot reach the DNA.

KEY IDEA

Acetylation tends to switch genes on by loosening chromatin. Methylation of DNA and of histones tends to switch genes off by tightening it. Both act on the same question: can the transcription machinery reach this gene?

Epigenetic control in human development: haemoglobin

Haemoglobin shows epigenetic control clearly. Adult haemoglobin contains two alpha and two beta globin chains. Fetal haemoglobin contains two alpha and two gamma globin chains. During development, the gamma genes are switched on first and then switched off just before and after birth, while the beta genes are switched on. The switch from gamma to beta production is not caused by any change in the genes themselves.

Histone acetylation appears to activate the fetal gamma genes. DNA methylation seems to silence them around birth, and histone methylation complements this by reinforcing the silencing. The graph shows how the proportion of each globin type changes with time from conception through to the months after birth.

EXAM TIP

When asked about epigenetic control in development, name the mechanism (acetylation or methylation), say what it does to chromatin, and link it to which gene is switched on or off at a given stage.

4. Phenotype, genotype and environment

A phenotype is the observable characteristic of an organism. It is not fixed by genotype alone. The genotype sets a potential, and the environment determines how much of that potential is reached. Height is the standard example: a person may carry genes for tall stature but still fail to reach full height if they are malnourished in childhood, because bone and muscle growth depend on protein and calcium. Health, which can also change gene expression, affects height in the same way.

Environmental effects can begin before birth. Poor maternal nutrition can leave a fetus short of nutrients needed for growth, and smoking reduces the oxygen supply, which can affect the growth of the body and brain. Twin studies help to separate genetic and environmental effects. Identical twins share essentially all their DNA, so differences between them are attributed to environment. Non-identical twins share about half their DNA on average, like ordinary siblings, but tend to share a more similar environment, which is why they are a useful comparison.

Skin and hair colour

Melanin is the dark pigment in skin and hair. It is produced by melanocytes, which are activated by melanocyte-stimulating hormone (MSH). MSH binds to receptors on melanocytes. Melanocytes package melanin into melanosomes, which are passed to surrounding skin and hair cells and gather around the nucleus, shielding the DNA from ultraviolet (UV) light. Individuals with more receptors and more melanin therefore have darker skin and greater protection against sunburn.

UV light increases the amount of MSH and the number of MSH receptors, making melanocytes more active. This is an environmental effect on a genotype-defined pathway. Hair can appear to lighten with UV exposure because UV damages melanin and other proteins in hair cells, even though more pigment is being made.

5. Continuous and discontinuous variation

Some characteristics are controlled almost entirely by genotype. Blood group is an example: the environment has little or no effect, and the phenotypes fall into separate categories with no overlap. Such characteristics are controlled by genes at a single locus and show discontinuous variation. A bar chart of their frequencies has gaps between the bars.

A characteristic controlled by one gene can be followed from parents to offspring with a genetic diagram such as a Punnett square.

Other characteristics, such as human height, show continuous variation. Their phenotypes form a smooth range with no clear categories. These characteristics are usually controlled by many genes at different loci, a pattern called polygenic inheritance, and they are also affected by the environment, either directly or by changing gene expression. When the frequency of each height category is plotted, the result is a bell-shaped curve, which indicates normally distributed data.

> **KEY IDEA**
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> Discontinuous variation: single locus, genotype dominant, discrete categories, no overlap. Continuous variation: many loci (polygenic), environment also matters, a range of values, normal distribution.

Polygenic inheritance

In polygenic inheritance, alleles at several loci interact to determine one characteristic. Each allele has only a small effect, and the combined effects of many alleles produce the individual's phenotype. Eye colour is a good example. Several loci influence how much pigment is in the iris. Brown eyes contain a lot of pigment, which absorbs light and makes them look dark. Blue eyes contain little pigment, so light is reflected from the iris. Intermediate pigment levels give the range of colours in between.

Adding more loci, and then the effect of the environment, turns a few separate classes into a bell-shaped curve.

QUICK CHECK

A characteristic shows continuous variation and a bell-shaped distribution. Which explanation is most likely?

- It is controlled by a single gene with two alleles and no environmental effect.
- It is controlled by many genes at different loci and is also affected by the environment.
- It is caused by a single mutation that changes the amino acid sequence of one protein.
- It is controlled by a single locus, and the phenotypes fall into separate groups.

Continuous variation with a normal distribution is produced by polygenic inheritance, where several loci each contribute a small effect, and the environment adds further variation. A single gene with no environmental effect gives discontinuous variation instead, which is the most common wrong answer.

QUICK CHECK

Which change would most directly switch a gene off by making its DNA less accessible to transcription?

- Acetylation of histones at the gene's region
- Demethylation of the cytosine bases in the gene
- Methylation of cytosine bases in the gene's DNA
- Splicing out an intron from the pre-mRNA

Methylation of cytosine next to guanine is linked to gene silencing, because the methyl group reduces transcription. Acetylation and demethylation both tend to activate genes. Splicing acts after transcription and does not change DNA accessibility, so it is the commonest wrong answer.

QUICK CHECK

A single gene produces three different proteins in three different cell types. What is the most likely reason?

- The gene sequence has mutated differently in each cell type.
- Each cell type has a different number of copies of the gene.
- Different combinations of exons are joined during splicing, giving different mRNAs.
- Each protein is produced by a different chromosome that carries the gene.

Alternative splicing joins the exons in different combinations, giving mRNAs of different length and sequence, and therefore different proteins. The DNA is the same in every cell, so mutation is not the explanation. Gene copy number is normally the same in each diploid cell.